

Statistics Review 1

Random Samples, Sampling Distributions, LLN and CLT

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Roadmap

1. From Probability to Statistics
2. Random Samples
3. The Sample Mean
4. The Sampling Distribution
5. The Law of Large Numbers
6. The Central Limit Theorem
7. Estimating the Population Variance
8. The Estimation Framework
9. Properties of Estimators
10. Summary

1. From Probability to Statistics

1.1 What Changes

Lectures 1 and 2 were **probability**: we assumed a distribution was *known* (“ $X \sim N(590, 25)$ ”) and asked questions about probabilities. **Statistics** flips the picture:

- We observe *data*: a sample of values drawn from some population.
- The underlying distribution is *unknown*.
- Our job is to make **inferences** about it — estimate its mean, its variance, test claims about it, predict future values.

Four words organize the rest of the course:

- **Population**: the random variable / distribution we care about.
- **Sample**: the observations we actually have.
- **Parameter**: a property of the population (unknown).

1.1 What Changes 2

- **Estimator**: a recipe that takes the sample and produces a guess at the parameter.

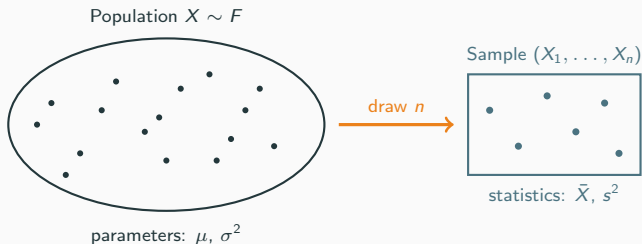
This lecture builds the toolkit. Lecture 4 uses it to construct confidence intervals and hypothesis tests.

2. Random Samples

2.1 Population vs Sample

Definition

The **population** is the underlying random variable X with distribution F (or density f). The **sample** is a collection $(X_1, X_2, \dots, X_n)$ of observations drawn from this distribution.



Key distinction. Population parameters (μ, σ^2) are *fixed unknowns*; sample statistics $(\bar{X}, s^2)$ are *random*, because they depend on which sample we drew.

2.2 i.i.d. Random Sample

Definition

A sample $(X_1, \dots, X_n)$ is an **i.i.d. random sample** from the distribution of X if it satisfies two conditions:

- **Identical**: each X_i has the same distribution as X , so $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2$ for every i .
- **Independent**: the X_i are mutually independent (Lecture 2 §5.3 extended to n variables).

Notation reminder. “i.i.d.” stands for “independent and identically distributed.” It is the standard sampling assumption in statistics. We met it informally in Lecture 1 §9.1 (Binomial as a sum of i.i.d. Bernoulli trials).

2.2 i.i.d. Random Sample 2

Formally, mutual independence means the joint distribution factorizes:

$$\mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n) = \prod_{i=1}^n \mathbb{P}(X_i \in A_i).$$

2.3 Real-World Caveats

Real data are often **not** truly i.i.d.:

- **Time series** (GDP, stock prices): today's value depends on yesterday's. Serial correlation breaks independence.
- **Cluster samples** (students within schools, workers within firms): observations in the same cluster share unmeasured characteristics. Within-cluster correlation breaks independence.
- **Network data**: connected units influence each other.
- **Selection effects**: not every member of the population is equally likely to appear in the sample.

The i.i.d. assumption is our **starting point** — the cleanest setting in which to derive results. Later courses (and later in this course) relax it.

Where this shows up later. Regression and resampling rely on independent observations. Regression also assumes $\mathbb{E}[\epsilon_i | X_i] = 0$ within each observation (Lectures 5–7).

3. The Sample Mean

3.1 Motivating Example: Average Height

Setup. You want to know the average height of adult men in the US — the **population mean** μ . You can't measure everyone, so you take samples of $n = 100$ men and compute the *sample mean* of each.

Three samples might give:

- Sample 1: $\bar{X} = 170$ cm
- Sample 2: $\bar{X} = 180$ cm
- Sample 3: $\bar{X} = 176$ cm

Two natural questions:

- **On average**, where does $\bar{X}$ land relative to the true μ ?
- **How spread out** is $\bar{X}$ across different samples?

This section answers both — and the answers are the most important formulas in statistics.

3.2 Definition of the Sample Mean

Definition

Let $(X_1, \dots, X_n)$ be an i.i.d. sample from a distribution with mean μ and variance σ^2 . The **sample mean** is

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Notation reminder. $\bar{X}$, pronounced “X-bar,” is standard notation for the average of a sample. It is itself a random variable: a different sample gives a different value of $\bar{X}$.

In the next two slides we ask: what is $\mathbb{E}[\bar{X}]$? And what is $\text{Var}(\bar{X})$? Both follow from the linear-combination rules of Lecture 2 §5.6 with $a_i = 1/n$.

3.3 Mean of $\bar{X}$ (Derived)

Key Result

For an i.i.d. sample with $\mathbb{E}[X_i] = \mu$,

$$\mathbb{E}[\bar{X}] = \mu.$$

Derivation. Apply linearity of expectation (Lecture 2 §5.6) with $a_i = 1/n$:

$$\begin{aligned}\mathbb{E}[\bar{X}] &= \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] && \text{(definition)} \\ &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] && \text{(linearity)} \\ &= \frac{1}{n} \cdot n\mu = \mu. && \text{(identical: } \mathbb{E}[X_i] = \mu\text{)}\end{aligned}$$

Interpretation. The sample mean is *centered* at the population mean. On average — across all possible samples — $\bar{X}$ hits μ exactly.

3.4 Variance of $\bar{X}$ (Derived)

Key Result

For an i.i.d. sample with $\text{Var}(X_i) = \sigma^2$,

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

Derivation. The X_i are independent, so apply the independent-case variance rule (Lecture 2 §5.6) with $a_i = 1/n$:

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) \quad (\text{definition})$$

$$= \sum_{i=1}^n \left(\frac{1}{n}\right)^2 \text{Var}(X_i) \quad (\text{independence; L2 §5.6})$$

$$= \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}. \quad (\text{identical: } \text{Var}(X_i) = \sigma^2)$$

3.4 Variance of $\bar{X}$ (Derived) 2

Interpretation. Variability shrinks with n . Bigger samples produce tighter estimates of μ .

3.5 The Standard Error

Take the square root of the variance of $\bar{X}$:

Key Result

$$\text{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

This quantity is called the **standard error** of the sample mean, written $\text{SE}(\bar{X})$.

Notation reminder. “Standard error” is the standard deviation of a *statistic*. We use SE rather than SD to emphasize that it refers to the sampling variability of $\bar{X}$, not to the standard deviation of the underlying population X .

The $1/\sqrt{n}$ rule. The standard error shrinks like $1/\sqrt{n}$ — so:

- To halve the standard error, you need $4\times$ as much data.

3.5 The Standard Error 2

- To shrink it by a factor of 10, you need $100\times$ as much data.

Diminishing returns: data is precious early, expensive later.

Where this shows up later. The standard error is the engine behind every confidence interval (Lecture 4) and every hypothesis test on a regression coefficient (Lecture 7).

3.6 Worked Example: Pizza Weights

Setup. A pizza chain sells large veggie pizzas with weight $W \sim N(24, 1)$ (mean 24 oz, SD 1 oz). You take a sample of $n = 10$ pizzas. What is $\mathbb{P}(\bar{W} < 23.5)$?

Step 1: distribution of $\bar{W}$. Each W_i is Normal, so $\bar{W}$ is Normal too (Lecture 2 §7.6: linear combinations of independent Normals are Normal):

$$\bar{W} \sim N\left(24, \frac{1}{10}\right), \quad \text{SE}(\bar{W}) = \frac{1}{\sqrt{10}} \approx 0.316.$$

Step 2: standardize and look up.

$$\mathbb{P}(\bar{W} < 23.5) = \mathbb{P}\left(Z < \frac{23.5 - 24}{0.316}\right) = \mathbb{P}(Z < -1.58) \approx 0.057.$$

3.6 Worked Example: Pizza Weights 2

Bonus question. What is $\mathbb{P}\left(\sum_{i=1}^{10} W_i < 235\right)$, the probability that the *total* weight of 10 pizzas is under 235 oz?

Same event, repackaged:

$$\sum W_i < 235 \iff \frac{1}{10} \sum W_i < 23.5 \iff \bar{W} < 23.5.$$

So $\mathbb{P}(\sum W_i < 235) = \mathbb{P}(\bar{W} < 23.5) \approx 0.057$.

Interpretation. Even a half-ounce shortfall in the per-pizza average is uncommon when you sample 10. The $1/\sqrt{n}$ rule at work.

4. The Sampling Distribution

4.1 The Sampling Distribution of $\bar{X}$

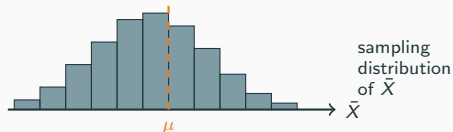
Every possible sample produces a different value of $\bar{X}$. Imagine *all* the samples you could have drawn, and collect the resulting $\bar{X}$ values into a distribution. That is the **sampling distribution of $\bar{X}$** .

Sample 0:  $\longrightarrow \bar{X}^{(0)} = 77.3$

Sample 1:  $\longrightarrow \bar{X}^{(1)} = 82.5$

Sample 2:  $\longrightarrow \bar{X}^{(2)} = 79.1$

$\vdots$



4.1 The Sampling Distribution of $\bar{X}$ 2

We already know two facts about this distribution from §3: it is centered at μ , and it has standard deviation $\sigma/\sqrt{n}$. Section 6.2 asks about its *shape*.

4.2 The Normal Case — Exact

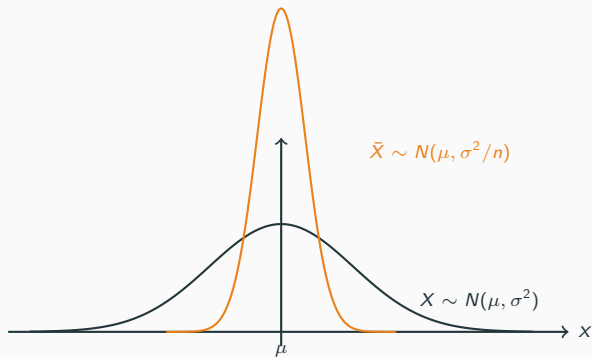
Key Result

If $X \sim N(\mu, \sigma^2)$ and $(X_1, \dots, X_n)$ is an i.i.d. sample, then

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{exactly, for every } n.$$

Why? This is a direct application of Lecture 2 §7.6 (any linear combination of independent Normals is Normal), with $a_i = 1/n$.

4.2 The Normal Case — Exact 2



Both curves are centered at μ . $\bar{X}$ has the same center but is *tighter* (smaller variance).

4.3 Worked Example: $\mathbb{P}(\bar{X} > 82)$

Setup. Population mean $\mu = 80$, variance $\sigma^2 = 36$. You draw an i.i.d. sample of $n = 36$. What is $\mathbb{P}(\bar{X} > 82)$?

(We don't need to assume X is Normal — $n = 36$ is comfortably large; we'll formalize this in §6 via the CLT.)

Step 1: standard error.

$$\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{6}{\sqrt{36}} = 1.$$

Step 2: standardize.

$$\mathbb{P}(\bar{X} > 82) = \mathbb{P}\left(Z > \frac{82 - 80}{1}\right) = \mathbb{P}(Z > 2) \approx 0.023.$$

Interpretation. If the population truly has $\mu = 80$, seeing a sample mean above 82 would happen in only about 2% of samples — enough evidence to be a little suspicious of the claim $\mu = 80$. This is the seed of hypothesis testing in Lecture 4.

5. The Law of Large Numbers

5.1 The Law of Large Numbers

Law of Large Numbers (LLN)

Let $(X_1, X_2, \dots)$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. As $n \rightarrow \infty$,

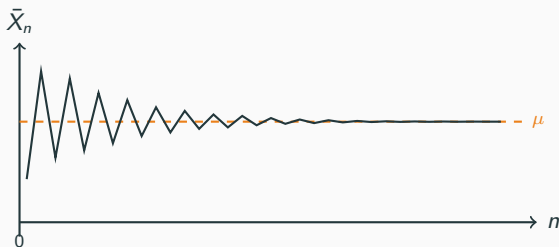
$$\bar{X}_n \xrightarrow{P} \mu.$$

The sample average converges (in probability) to the population mean.

Notation reminder. $\bar{X}_n \xrightarrow{P} \mu$ reads “ $\bar{X}_n$ converges in probability to μ .”

Informally: the probability of $\bar{X}_n$ being more than any fixed distance from μ goes to zero as n grows.

5.1 The Law of Large Numbers 2



Each step adds one more observation to the running average. Wild at small n , settling toward μ as n grows.

5.2 LLN as the Foundation of Statistics

Why does the LLN matter? Because it is the formal reason that **collecting more data works**.

Without LLN, there would be no guarantee that $\bar{X}$ tells you anything about μ — even an enormous sample might give a wildly wrong answer. LLN rules this out: the sample average is essentially certain to be close to the population mean once n is large.

Compare to §3.3. We showed $\mathbb{E}[\bar{X}] = \mu$: *on average* $\bar{X}$ equals μ . LLN is stronger: it says for any *single* large sample, $\bar{X}$ is close to μ with high probability.

Where this shows up later.

- LLN is what makes the sample mean a **consistent** estimator of μ — formal definition in §9.3.
- It is also the foundation of Monte Carlo simulation and bootstrap resampling (Lecture 13): averages of independent simulated values converge to the quantity you want to estimate.

6. The Central Limit Theorem

6.1 What If the Population Isn't Normal?

In §4.2, $\bar{X}$ was exactly Normal — but only because we *assumed* the population X was Normal. What if it isn't?

- Income, wealth: right-skewed, heavy upper tail.
- Stock returns: heavy tails on both sides.
- Counts of customers, calls: skewed, often Poisson-like.

Surprisingly, the answer is:

$\bar{X}$ is still approximately Normal for large n , regardless of the shape of X .

This is the most celebrated result in probability theory — the **Central Limit Theorem**.

6.2 The Central Limit Theorem

Central Limit Theorem (CLT)

Let $(X_1, X_2, \dots, X_n)$ be i.i.d. with mean μ and finite variance σ^2 . Then for large n ,

$$\bar{X} \overset{\text{approx}}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right),$$

or equivalently, the standardized sample mean converges in distribution to a standard Normal:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1) \quad \text{as } n \rightarrow \infty.$$

Notation reminder. $\xrightarrow{d}$ reads “converges in distribution to.” Don’t sweat the formal definition; the picture on the next slide is what to remember.

Compare LLN and CLT. LLN tells you *where* $\bar{X}$ ends up ($\rightarrow \mu$). CLT tells you the *shape* of its fluctuations around μ (Normal, width $\sigma/\sqrt{n}$).

6.3 The CLT in Three Pictures

Roll a fair six-sided die n times and average the results. The population is uniform on $\{1, 2, 3, 4, 5, 6\}$ — definitely not Normal. Below: simulated histograms of $\bar{X}$ for three sample sizes.



$n = 1$: a single die roll — flat distribution. $n = 5$: averaging starts to concentrate around 3.5 and develop a bell shape. $n = 30$: practically indistinguishable from a Normal.

6.4 Rule of Thumb and Caveats

How big does n need to be for the CLT approximation to be good?

Common rule of thumb: $n \geq 30$.

But this is not a magic number. The required n depends on the shape of the population:

- Mildly skewed: $n = 30$ is often plenty.
- Heavily skewed or heavy-tailed (incomes, financial returns): you may need n in the hundreds.
- Symmetric and near-Normal already: the approximation is good even for small n .

6.4 Rule of Thumb and Caveats 2

Caveat: finite variance. The CLT requires $\text{Var}(X_i) < \infty$. Distributions with infinite variance (some heavy-tailed models in finance) violate this and the CLT does *not* apply.

Where this shows up later. The CLT is the formal license to use Normal-based confidence intervals (Lecture 4) and hypothesis tests (Lecture 7) even when we don't know the population shape. *A note for L4:* when we additionally substitute s for σ in the standard error, we use a stricter rule of thumb $n \geq 60$ — see Lecture 4 §3.2.

6.5 Worked Example: Unknown Population

Setup. A population has unknown shape, but you know its mean $\mu = 90$ and standard deviation $\sigma = 15$. You take an i.i.d. sample of size $n = 25$. What is $\mathbb{P}(85 < \bar{X} < 92)$?

Step 1: standard error.

$$\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{25}} = 3.$$

Step 2: invoke the CLT. With $n = 25$ and a not-too-wild population, the CLT gives

$$\bar{X} \overset{\text{approx}}{\sim} N(90, 3^2).$$

6.5 Worked Example: Unknown Population 2

Step 3: standardize and look up.

$$\begin{aligned}\mathbb{P}(85 < \bar{X} < 92) &= \mathbb{P}\left(\frac{85 - 90}{3} < Z < \frac{92 - 90}{3}\right) \\ &= \mathbb{P}\left(-\frac{5}{3} < Z < \frac{2}{3}\right) \\ &= \Phi(0.67) - \Phi(-1.67) \\ &\approx 0.748 - 0.048 = 0.700.\end{aligned}$$

Interpretation. About 70% of samples have means in $(85, 92)$ — and we did this calculation *without ever specifying the shape of the population*. That's the power of the CLT.

7. Estimating the Population Variance

7.1 The Sample Variance

In practice, the population variance σ^2 is usually unknown. We estimate it from the sample.

Definition

The **sample variance** of an i.i.d. sample $(X_1, \dots, X_n)$ is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

The **sample standard deviation** is $s = \sqrt{s^2}$.

Why $n-1$, not n ? With this denominator, s^2 is an *unbiased* estimator of σ^2 : $\mathbb{E}[s^2] = \sigma^2$.

Intuition: we plug in $\bar{X}$ (estimated from the data) in place of the true μ . Because $\bar{X}$ is itself chosen to be close to the X_i , the squared deviations

7.1 The Sample Variance 2

$(X_i - \bar{X})^2$ are systematically a bit *smaller* than they would be against the true μ . Dividing by $n - 1$ instead of n corrects for this. The number $n - 1$ is called the *degrees of freedom*.

(Full derivation of $\mathbb{E}[s^2] = \sigma^2$ is in any standard stats text; we'll take it as given.)

7.2 The Estimated Standard Error

When σ is unknown, replace it by s in the standard-error formula:

Estimated standard error

$$\hat{SE}(\bar{X}) = \frac{s}{\sqrt{n}}.$$

This is the version of the standard error you'll actually use in practice: σ is rarely known, but s can always be computed from the sample.

A subtle consequence. When we substitute s for σ , the standardized statistic

$$\frac{\bar{X} - \mu}{s/\sqrt{n}}$$

7.2 The Estimated Standard Error 2

no longer follows $N(0, 1)$ exactly — because s itself is random. In a traditional stats course you would meet the **Student's t distribution** at this point.

Where this shows up later. We will instead take a **large-sample approach** in Lecture 4: for $n \geq 60$, the statistic $(\bar{X} - \mu)/(s/\sqrt{n})$ is approximately $N(0, 1)$ to high accuracy, even with s in place of σ . This keeps all the inference machinery in this course on a single distribution — the Normal.

8. The Estimation Framework

8.1 Parameter, Statistic, Estimator

Three words that are easy to confuse:

- **Parameter:** a fixed (unknown) property of the population — e.g., μ , σ^2 , p . *Not* random; it is just unknown.
- **Statistic:** any function of the sample — e.g., $\bar{X}$, s^2 , the sample maximum. *Random*, because it depends on the sample.
- **Estimator:** a statistic *used as a guess* for a specific parameter. We write $\hat{\theta}$ for an estimator of a parameter θ .

Notation reminder. The “hat” on $\hat{\theta}$ marks it as an estimator. So $\hat{\mu} = \bar{X}$ says: “our estimator of μ is $\bar{X}$.”

Examples we already know:

- $\hat{\mu} = \bar{X}$ estimates μ .
- $\hat{\sigma}^2 = s^2$ estimates σ^2 .

The question for §9: what makes an estimator *good*?

8.2 Worked Example: A Tiny Population

Setup. Imagine a tiny “population” of four students with scores

$$\{75, 80, 85, 90\}.$$

Population mean: $\mu = (75 + 80 + 85 + 90)/4 = 82.5$.

Step 1: list all samples of size 2. There are $\binom{4}{2} = 6$ unique pairs.

$$(75, 80) : \bar{X} = 77.5$$

$$(75, 85) : \bar{X} = 80.0$$

$$(75, 90) : \bar{X} = 82.5$$

$$(80, 85) : \bar{X} = 82.5$$

$$(80, 90) : \bar{X} = 85.0$$

$$(85, 90) : \bar{X} = 87.5$$

8.2 Worked Example: A Tiny Population 2

Step 2: average the six sample means.

$$\frac{77.5 + 80 + 82.5 + 82.5 + 85 + 87.5}{6} = \frac{495}{6} = 82.5.$$

Interpretation. The average of all possible sample means equals the population mean exactly. This is **unbiasedness**, $\mathbb{E}[\bar{X}] = \mu$, made concrete with finite arithmetic.

Where this shows up later. The same logic underlies every “ $\bar{X}$ is unbiased for μ ” argument in the rest of the course.

9. Properties of Estimators

9.1 Unbiasedness

Definition

An estimator $\hat{\theta}$ is **unbiased** for θ if

$$\mathbb{E}[\hat{\theta}] = \theta.$$

Across all possible samples, on average $\hat{\theta}$ lands on θ .

Examples we have already shown:

- $\bar{X}$ is unbiased for μ (§3.3).
- s^2 is unbiased for σ^2 (§7.1) — this is why the denominator is $n - 1$.

Caution. Unbiased $\neq$ “close to the truth.” An unbiased estimator can still have huge variance — it just hits the truth *on average*. We need a second property to capture “tightness.”

9.2 Efficiency

Definition

Among unbiased estimators of θ , an estimator $\hat{\theta}$ is **more efficient** than a competitor $\tilde{\theta}$ if

$$\text{Var}(\hat{\theta}) < \text{Var}(\tilde{\theta}).$$

The most efficient estimator has the smallest variance among unbiased ones.

Efficiency captures *tightness*: a more efficient estimator has its estimates clustered more tightly around the truth.

Where this shows up later. Comparing OLS to alternative regression estimators (Lecture 7); regularization deliberately introduces *bias* to gain efficiency (Lecture 10).

9.3 Consistency

Definition

An estimator $\hat{\theta}_n$ is **consistent** for θ if

$$\hat{\theta}_n \xrightarrow{P} \theta \quad \text{as } n \rightarrow \infty.$$

With enough data, the estimator gets arbitrarily close to the truth (with high probability).

By the Law of Large Numbers (§5), $\bar{X} \xrightarrow{P} \mu$. So $\bar{X}$ is **consistent for μ** .

Consistency is the *bare minimum* we ask of an estimator: if collecting more data doesn't help, the estimator is broken. Unbiased estimators can still be inconsistent (see the next slide) and biased estimators can be consistent — the two properties are distinct.

9.4 Worked Example: $\bar{X}$ vs. $\tilde{X}$

Setup. You have an i.i.d. sample $(X_1, \dots, X_{100})$ with mean μ and variance σ^2 . Consider two estimators of μ :

- $\bar{X} = \frac{1}{100} \sum_{i=1}^{100} X_i$ (use all observations).
- $\tilde{X} = (X_1 + X_2)/2$ (use only the first two).

Step 1: unbiasedness.

$$\mathbb{E}[\tilde{X}] = \frac{\mathbb{E}[X_1] + \mathbb{E}[X_2]}{2} = \frac{\mu + \mu}{2} = \mu = \mathbb{E}[\bar{X}].$$

Both estimators are unbiased.

9.4 Worked Example: $\bar{X}$ vs. $\tilde{X}$ 2

Step 2: variance.

$$\text{Var}(\tilde{X}) = \frac{\text{Var}(X_1) + \text{Var}(X_2)}{4} = \frac{2\sigma^2}{4} = \frac{\sigma^2}{2},$$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{100}.$$

Step 3: efficiency. $\text{Var}(\bar{X}) / \text{Var}(\tilde{X}) = 1/50$. The sample mean is **50 times more efficient** than $\tilde{X}$ at $n = 100$.

Step 4: consistency. As n grows:

- $\text{Var}(\bar{X}) = \sigma^2/n \rightarrow 0$ as $n \rightarrow \infty$. So $\bar{X}$ is **consistent**.
- $\text{Var}(\tilde{X}) = \sigma^2/2$ for every n — it doesn't depend on the sample size at all. So $\tilde{X}$ is **not consistent**; collecting more data doesn't help.

9.4 Worked Example: $\bar{X}$ vs. $\tilde{X}$ 3

Interpretation. Two estimators can both be unbiased and still differ enormously in quality. Using all the data matters.

Where this shows up later. In Lecture 10 (regularization), we'll see estimators that are deliberately *biased* but smaller-variance than OLS — the bias–variance tradeoff.

10. Summary

10.1 Take Away — Foundations

What's new in statistics

- **Population**: random variable with unknown distribution (parameters $\mu, \sigma^2, \dots$).
- **Sample**: i.i.d. observations $(X_1, \dots, X_n)$. i.i.d. = identical distribution + mutual independence.
- **Parameter** (fixed but unknown) vs. **statistic** (random function of the sample) vs. **estimator** (statistic used to guess a parameter, written $\hat{\theta}$).

10.2 Take Away — Sample Mean Properties

Exam targets

For i.i.d. $(X_1, \dots, X_n)$ with $\mathbb{E}[X_i] = \mu$, $\text{Var}(X_i) = \sigma^2$:

- **Definition:** $\bar{X} = \frac{1}{n} \sum_i X_i$.
- **Mean:** $\mathbb{E}[\bar{X}] = \mu$ (unbiased).
- **Variance:** $\text{Var}(\bar{X}) = \sigma^2/n$.
- **Standard error:** $\text{SE}(\bar{X}) = \sigma/\sqrt{n}$. Shrinks like $1/\sqrt{n}$.
- **Sample variance:** $s^2 = \frac{1}{n-1} \sum_i (X_i - \bar{X})^2$, unbiased for σ^2 .
- **Estimated SE:** $\hat{\text{SE}}(\bar{X}) = s/\sqrt{n}$ when σ is unknown.

10.3 Take Away — LLN and CLT

Two big results

- **Law of Large Numbers:** $\bar{X}_n \xrightarrow{P} \mu$. The sample mean converges to the population mean. *Tells you where $\bar{X}$ ends up.*
- **Central Limit Theorem:** for large n ,

$$\bar{X} \stackrel{\text{approx}}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right),$$

regardless of the population shape. *Tells you the shape of its fluctuations.*

Together: LLN says the average is going to the right place; CLT says how it wobbles around that place. The CLT is what makes Normal-based inference work even when the underlying population is not Normal.

10.4 Take Away — Estimator Properties

Three properties of estimators

- **Unbiased:** $\mathbb{E}[\hat{\theta}] = \theta$.
- **Efficient:** smaller $\text{Var}(\hat{\theta})$ among unbiased estimators.
- **Consistent:** $\hat{\theta}_n \xrightarrow{P} \theta$ as $n \rightarrow \infty$.

Punchline from §9.4: $\bar{X}$ and $\tilde{X} = (X_1 + X_2)/2$ are both unbiased estimators of μ , but $\bar{X}$ is much more efficient (σ^2/n vs. $\sigma^2/2$) and is the only one of the two that is consistent. Using all the data matters.

10.5 Where Each Tool Reappears in This Course

Forward pointers

- **Standard error** $\sigma/\sqrt{n}$ and $s/\sqrt{n}$ — confidence intervals (Lecture 4); large-sample tests on regression coefficients (Lecture 7).
- **Sample variance** s^2 — estimated standard errors for large-sample inference (Lecture 4); OLS variance estimation (Lecture 7).
- **CLT** — Normal-based inference even when the population isn't Normal (Lecture 4); OLS asymptotic theory (Lecture 7).
- **LLN** — consistency of OLS (Lecture 6); foundation of bootstrap and Monte Carlo (Lecture 13).
- **Estimator properties** — bias–variance decomposition and regularization tradeoffs (Lecture 10); maximum likelihood (Lecture 12).

Next time: confidence intervals and hypothesis tests — how to translate the sample mean into statements about μ .